

Table 5: NNE in Search Model with Alternative Moment Specifications

Number of moments	Total bias	RMSE
16	1.210 (.027)	0.649 (.017)
32	1.201 (.026)	0.630 (.015)
40	0.914 (.029)	0.508 (.015)
46	0.183 (.029)	0.311 (.017)
60	0.186 (.035)	0.291 (.015)
81	0.171 (.033)	0.279 (.015)

Notes: Each row corresponds to one specification of moments. Total |bias| is the sum of absolute biases across the parameters of the search model. Results are based on 100 Monte Carlo datasets. Numbers in parentheses are standard errors.

4.5 Additional results on search model

We provide further results on the performance of NNE in estimating the consumer search model. In particular, we examine the impacts of moment choice and data size.

Choice of moments

We measure how the estimation accuracy of NNE varies as the moment choice varies. This exercise is related to the general discussion in Section 2 on NNE’s robustness to redundant moments. (Also see Section 3 for a similar exercise under the much simpler AR(1) model.)

Recall that so far we have included 46 data moments in $\mathbf{m}$ (see Section 4.3). We now construct five alternative specifications of $\mathbf{m}$ by removing or adding moments. The first specification removes the covariance matrix of $\tilde{\mathbf{y}}_i$ (40 moments remain). The second specification further removes moments by dropping the non-free search dummy from $\tilde{\mathbf{y}}_i$ (32 moments remain). The third specification removes $\tilde{\mathbf{y}}_i$ all together (16 moments remain). The fourth specification adds to the original 46 moments the cross-covariances between $\mathbf{y}_{ij}$ and $\mathbf{x}_{ij}^2$ (60 moments total). The fifth specification further adds the cross-covariances between $\tilde{\mathbf{y}}_i$ and $J^{-1} \sum_{j=1}^J \mathbf{x}_{ij}^2$ (81 moments total).

Table 5 reports the estimation accuracy under the six specifications of moments, with each row corresponding to one specification. The first column lists the numbers of moments. We do not see sizable increases in bias or RMSE as we move from 16 to 81 moments. Instead, the bias and RMSE largely decrease. The result suggests that NNE is able to capitalize on the information from additional moments while being robust to potentially redundant moments.

Data size

Our analysis so far has focused around data size $n = 1000$. We now examine how the estimation accuracy of NNE varies with data size and compare it to SMLE. To facilitate the comparison, we keep the simulation burdens about the same between NNE and SMLE (by using appropriate L^* and R).

Figure 9 reports the results for $n \in \{500, 1000, 2500, 5000\}$. The two plots differ in simulation burden, and a higher burden is used in the right plot. Within each plot, NNE and SMLE are